

Name:Supriya.B Register Number:250701760 Experiment name: Linear Regression

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

YearsExperience Salary

```
0 1.1 39343
1 1.3 46205
2 1.5 37731
3 2.0 43525
4 2.2 39891
5 2.9 56642
6 3.0 60150
7 3.2 54445
8 3.2 64445
9 3.7 57189
10 3.9 63218
11 4.0 55794
12 4.0 56957
13 4.1 57081
14 4.5 61111
15 4.9 67938
16 5.1 66029
17 5.3 83088
18 5.9 81363
19 6.0 93940
20 6.8 91738
21 7.1 98273
22 7.9 101302
23 8.2 113812
24 8.7 109431
25 9.0 105582
26 9.5 116969
27 9.6 112635
28 10.3 122391
29 10.5 121872
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
# Column Non-Null Count Dtype
---
0 YearsExperience 30 non-null float64
1 Salary 30 non-null int64
```

```
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
df.dropna(inplace=True)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
# Column Non-Null Count Dtype
-----
0 YearsExperience 30 non-null float64
1 Salary 30 non-null int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.describe()
```

```
YearsExperience Salary
count 30.000000 30.000000
mean 5.313333 76003.000000
std 2.837888 27414.429785
min 1.100000 37731.000000
25% 3.200000 56720.750000
50% 4.700000 65237.000000
75% 7.700000 100544.750000
max 10.500000 122391.000000
```

```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=42)
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

```
LinearRegression()
```

```
model.score(x_train,y_train)
```

```
0.9645401573418146
```

```
model.score(x_test,y_test)
```

```
0.9024461774180497
```

```
model.coef_
```

```
array([[9423.81532303]])
```

```
model.intercept_
```

```
array([25321.58301178])
```

```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

```
print("Estimated Salary for {} years of experience is {}: "
      .format(yr_of_exp,Salary))
```

```
Estimated Salary for 44.0 years of experience is [[439969.45722514]]:
```